

- 6 Frictional force on the masses provides the centripetal force for their motion

$$F_c = f = mr\omega^2$$

$$f_{3kg} = (3.0)(2.0)\omega^2$$

$$f_{6kg} = (6.0)(5.0)\omega^2$$

$$\frac{f_{3kg}}{f_{6kg}} = \frac{6.0}{30} = 0.20$$

Answer: A